LINKING ANALYSIS AND CREATION: AN IMPROVISED STUDY

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ABSTRACT

This paper discusses the application of a compositional process linked to the analysis of specific improvised musical works incorporating new technological elements. A parametric analysis template is utilized to examine nine works and insights from the analyses are applied to the creation of an improvised study. The paper draws from doctoral research where an analysis/creation model was developed and applied across a range of musical genres including electronic, electronic dance music, improvised and rock.

INTRODUCTION

Hill (2007) reviewed a range of existing analytical methods and identified the tendency for analysis to be directed towards musicological, rather than music making, ends. Furthermore, many existing analytical methods are inadequate in dealing with the processes and outputs of new and emerging music technologies. Existing analytical methods focusing on electroacoustic, electronic dance music, jazz and popular music provide useful analytical tools. Of particular importance is Middleton's (2000) notion of a *participant analyst*. It is the premise of this research that extending the role of Middleton's *participant analyst* beyond that of the musicological domain, to encompass a *participant analyst/creator*, has the potential to facilitate a more comprehensive investigation into the compositional/musical creation domain.

ANALYTICAL METHOD

The text-based, parametric approach incorporates observations made by the analyst (mostly on an aural basis) alongside insights from the literature related to particular parameters. The first section of the analysis template (see Table 1) addresses the shaping factors in the creation of a musical work. According to numerous authors (e.g., Brown, 1997; Ferrara, 1984; Hubbs, 2000; Tagg, 2000), the consideration of a broad range of extramusical parameters is necessary for any comprehensive analysis this research incorporates this principle. The consideration of shaping factors also enables an investigation of positionality and individual agency in relation to technology. The second section of the template addresses the inputs of a particular work and includes a range of traditional parameters (e.g., pulse, metre, dynamics, pitch selection, texture, etc.) in addition to parameters more suited to electronic and electroacoustic works such as, sound objects, spatial elements, and programmatic association. The range of parameters is intended to include those that impact on the creative process.

Shaping Factors:

Parameter			
Musical	**Theoretical**		*Philosophical, music theoretical.*
	Practical	**Technical**	*Tools for realisation e.g., equipment, studios (constraints/potential).*
		Play	*Bottom up - type approaches, e.g., jamming, software/synthesiser exploration.*
		Practice	*Instrumental/vocal/studio etc.*
	Listening		*Influences, musical or other.*
Other	**Environmental**	**Macro**	*Time, place, culture etc.*
		Micro	*Room, ambience, etc.*
	Budget/ Resources		*Source of funds, amount.*
	Intended Audience		*Explicitly stated? Implications due to marketing?*
	Timeframe		*Time for realisation.*

Inputs:

Parameter		Experiential	Literature
Sound	**Sources**	*E.g., electronic, concrète, instrumental, vocal.*	
	Objects	*Individual elements – e.g. durations/amplitude envelopes, frequency ranges/pitches, wave types, description of samples, text fragments, composite objects (e.g., additive synthesis). (Roads, 2001, following Schaeffer)*	
	Object processing	*Types, e.g., delay, chorus, reverb, compression, gate, filtering, distortion, speed variation, LF modulation.*	
Rhythm	**Pulse**	*Present? Constant/variations? Tempo – constant/varied, sudden or gradual changes? (Solomon, 2002).*	
	Metre	*Ametric, polymetric, multimetric? Regular meter? Constant or changing? (Solomon, 2002)*	
	Patterning	*Description of patterns, changes, note values, accents, effects on other parameters (e.g. pitch) (Solomon, 2002).*	
Pitch	**Selection**	*Tonal, atonal, microtonal, polytonal, modal, chromatic, aleotoric? (Adapted from Solomon - "Tonality", 2002)*	
	Vertical structures	*Chord structures, voicings, intervals (Solomon – "Harmony", 2002).*	
	Vertical patterning	*Chord sequence/repetition/ variation/ rate of change/ root movement/ pedal point/drone. (Solomon - "Harmonic Motion", 2002)*	
	Horizontal structures and patterning	*Melody/Phrase/motive/riff – structure/contour/range/length. Description of patterns. Repetition/variation? (Adapted from Solomon - "Thematic/motivic structure", "Thematic/motivic development" and "Pitch range", 2002).*	
	Dynamics	*Constant/Changing? Independent of texture? (Solomon, 2002)*	
	Texture	*Monophonic, homophonic, polyphonic, contrapuntal, heterophonic? Homorhythmic? Thin/thick? Changing? Polarization (melody/accompaniment, solo/tutti, antiphony. etc.) (Solomon, 2002).*	
	Timbre	*Quality(ies). Static/evolving. Sound combinations/blending (Solomon, 2002). Vocal timbre – throat/head/chest and gender construction (Shepherd, 1991, p. 163).*	
	Spatial elements	*Nature of perceived acoustic space. (Wishart, 1996, p. 140) Stereo - left/right, near/far, static/dynamic, local/diffuse (Wishart, 1996, ch. 10). Reference to "Sound box" (Moore, 1993, p. 106) Multispeaker – placement, static/dynamic, trajectories, local/diffuse (Wishart, 1996, ch. 10).*	
	Programmatic Association	*Perceived meaning/connotation. Implicit/explicit, Intrinsic/contextual recognition (Wishart, 1996, p. 150). Sound image as metaphor? (Wishart, 1996, p. 165) Reference to physical gesture? (Middleton, 2000, p. 108) Signification/coding (Middleton, 1990, ch. 6). Syntax/semantics/ontological levels (Ferrara 1984, p. 359).*	
	Structure	*Form type(s). Small and large scale relationships. Derivation of structure – principle? (Solomon, 2002).*	
	Interaction	*Description. Rules? Standard types of interaction? (Rinzler, 1988)*	
	Score	*Existing? Type?*	
	Presentational Format	*Recording (format?)/ Live performance (venue?)*	

Table 1. Analysis Template

COMPOSITIONAL METHOD

A compartmentalized and hybridized approach to composition is adopted whereby various composition practices are identified during analysis forming a storehouse from which new compositional work is undertaken. The relationships between compositional methods identified via analysis and those utilised in the creation of new works occur on a range of levels and are particular to the subjective position of my role as participant analyst/creator. At the shaping factors level, some parameters are beyond my control. These include certain *technical, environmental, budgetary* and *timeframe* considerations. By framing my position at the shaping factors level in terms of resonance, links can be made both between works analysed and to the new creative works developed. For example, whilst not having the specific technical limitations involved in a particular work, the general notion of 'technical limitation' affords a point of departure for a discussion of such in the creative process. Similarly, the comparison of the shaping factors between works and to my own position offers the possibility of insight into aspects of continuity and/or disruption both within and across musical genres.

Some inputs can be directly incorporated into new works whereas others can be utilised at a more general conceptual level. For example, the use of a quadraphonic spatial arrangement can be directly transferred to a new work, whereas a concept such as the absence of a gestural connotation for the generation of sounds in an electroacoustic work can be transferred at a general level, regardless of the particular sound utilised in a new work. Similarly, certain inputs identified in the analysed works may provide a point of departure for the development of inputs for new works. The development of integral serialism in the 1940s and 1950s, whereby serial procedures were applied to a range of parameters (e.g., rhythm and dynamics) beyond pitch, provides one such example.

WORKS ANALYSED

The selected works analysed are listed below:

1. Pauline Oliveros: "Bye Bye Butterfly" (1965)
2. Miles Davis: "Bitches Brew" (1969)
3. Herbie Hancock: "Chameleon" (1974)
4. David Behrman: "On the Other Ocean" (1978)
5. Joel Chadabe: "Valentine" (1987 – 94)
6. George Lewis: "Voyager" (1987 – 95)
7. Courtney Pine: "Oneness of Mind" (1997)
8. *Interface*: "Scrb" (2000)
9. Dave Douglas: "November" (2003)

APPLYING ANALYTICAL METHODOLOGY

Completed analyses of improvised works are presented in Hill (2007). Most of the works analysed are recordings of live performances, with the exception of "Bitches Brew", "Chameleon", "Oneness of Mind" and "November" where some degree of overdubbing and/or editing of performance has occurred in the production of the recording. The limitations of a desk-based (i.e., using the recording as analysis object) approach to analysis include Butterfield's (2002) objection to the lack of context, or in the case of jazz performance, the 'carnival atmosphere' surrounding the event. Given that most of the works analysed were recorded live in a studio, not in front of an audience (except "Voyager" and "Scrb"), the 'carnival' is somewhat restricted to the happenings within the studio.

The reliance on a recording for analysis presents difficulties in the analysis of interaction. Firstly, the ability to identify sound source with a performer is crucial for any discussion of interpersonal/machine interaction. However, in the case of Chadabe's "Valentine", the situation is further complicated because the composer deliberately sought to obscure the differentiation between the computer and human performances. Secondly, in the case of ensemble interaction, visual cues can provide insight into levels of interaction that may not be apparent aurally.

It is possible to group the nine improvised works into two sub-groups: (a) works emerging from a 'jazz' tradition (including "Bitches Brew", "Chameleon", "Oneness of Mind", and "November"), and (b) works emerging from an 'art music' tradition (including "Bye Bye Butterfly", "On the Other Ocean", "Voyager", "Valentine" and "Scrb"). This division is made primarily on the basis of the stylistic history of the artist and is useful in the consideration of genre terrain.[1]

ASCERTAINING COMPOSITIONAL METHODS/ MAPPING GENRE TERRAIN

In terms of *shaping factors*, a somewhat diverse range impacts upon the creation of the improvised works analysed herein. At a *theoretical* level, one broad similarity between the composers/performers of the works is university or conservatorium study. Whilst most citations in the literature of such study pass unremarkably, neither Davis nor Oliveros view such study as particularly instrumental in developing their own careers. Oliveros, for example, identifies the disjunction between her own creativity and the instruction occurring in academia:

> I had a struggle for years fending off the structures that were being brought forward by instructors in academia … they had no relationship to what I was hearing. I resisted following the instructors' models.

[1] I acknowledge the problematic nature of this division, particularly the blurring of such boundaries by, for example, George Lewis' involvement with many artists working in a 'jazz' tradition, and also Joel Chadabe's album of reworked 'standards' - *After Some Songs* (1987-94).

> Somehow or other, the listening inwardly created the space to go ahead and the courage to do what I felt was important to do (Oliveros, 1993, p. 377).

Davis echoes this disjunction, withdrawing from the Julliard School of Music as he was finding more value in playing with and listening to New York jazz artists (Carr, 1998).

The need for some degree of performer autonomy is prevalent amongst the composers/performers of the improvised works. Remarks to this effect are found in the *theory*, *play* and/or *practice* parameters of the analyses. The notion of a 'shared creation' is posited by Behrman (1997) as a contrast to the composer/performer division common to the European art music tradition.

> There's the model especially in the European tradition of the Creative Superperson (the Composer), and the lesser worker musician (the performer) which I've wanted to get away from. I like the idea of sharing in the creation of something and don't mind getting less than 100% of the credit for it. I like designing software which can be lifted off the ground, so to speak, by a wonderfully imaginative musician who does something with it that I never would have dreamed of… The tradition of 'unfinished composition' of course is not new. Much of Jazz and other musics primarily designed for live performance have a lot to do with that kind of idea. You could say that when the composition is unfinished, authority is being questioned (Behrman cited in Gross, 1997).

Chadabe suggests that he composes 'activities' rather than 'pieces':

> A 'piece', whatever its content, is a construction with a beginning and end that exists independent of its listeners and within its own boundaries of time. An 'activity' unfolds because of the way people perform; and consequently, an activity happens in the time of living; and art comes closer to life (n.d.).

The degree to which performer autonomy is embedded within each of the improvised works discussed herein varies and appears to correlate with the extent to which such autonomy is discussed by the artists themselves in the literature. For example, in "Voyager", Lewis is explicitly interested in creating a non-hierarchical environment where the computer 'performer' displays a similar degree of autonomy to that of the human performer. On the other hand, in comments from Davis, from musicians playing with him, and from those influenced by him (e.g., Douglas), a picture emerges of an artist ('creative superperson?') whose whole 'musical conception' (Gitler, 2002) included the careful selection of side-people and musical materials alongside a "knack of pulling things out of musicians that they might not normally be aware of" (McLaughlin cited in Carr, 1998, p. 263-4). Thus although the performance of Davis' ensemble was experimental and improvised, there is a sense that Davis was in control over the direction of the sonic outcomes. This is at odds with both the non-hierarchical approach of Lewis and the clear delineation of roles outlined by Chadabe and Behrman above.

In addition to a range of contemporary musical and extra-musical influences, many improvising musicians also explore the potential of new technologies. Amongst the works analysed for this study, the rationale for, and nature of that exploration are quite disparate. Behrman and Lewis present a similar attitude towards the development of interactive computer works. For example, for Behrman, technology as such is amoral and dependent on the motivation of the user/developer (Behrman cited in Gross, 1997). Lewis (2000b) echoes this by linking broader social and cultural structures to software development:

> Musical computer programs, like any texts, are not 'objective' or 'universal', but instead represent the particular ideas of their creators. As notions about the nature and function of music become embedded into the structure of software-based musical systems and compositions, interactions with these systems tend to reveal characteristics of the community of thought and culture that produced them (p. 33).

Interaction between sound sources and/or performers occurs on a range of levels and via a range of parameters. A solo and accompaniment model is common to the jazz tradition works ("Bitches Brew", "Chameleon", "Oneness of Mind" and "November"). However the degree to which the accompaniment instruments are 'locked' to particular parts varies. In contrast to the solo/accompaniment model, collective improvisation involving all or some of the sound sources/performers is an aspect of most of the works and features most prominently in "Voyager" and "Scrb".

Rules for interaction are established in the interactive computer works and embed varying degrees of human control. Chadabe uses the software application M (which he developed), where basic MIDI information is recorded and then pitch, rhythm and timbre are manipulated by collections of algorithms accessed via graphic display by the user (Chadabe, 1997). The applications utilised by Behrman and Lewis operate in real-time without direct human control and are based on the computer 'listening' to the human performance and responding in certain ways. For example, the "Voyager" system, when multiple performers are present, decides which performer to listen to and whether to match, oppose or ignore various parameters (Lewis, 2000b). The development of such rules in some way represents a 'top down' process of creation. However, the sonic outcomes of such systems are determined by the performers and hence represent, simultaneously, a 'bottom up' approach. The traditional model of playing jazz standards, (i.e., the

model used for many of the works considered here) where a soloist improvises over a set form, represents a similar merging of 'top down' and 'bottom up' processes.

APPLYING COMPOSITIONAL METHODS/ DEVELOPMENT OF IMPROVISED STUDY

Particular *resonances* observed in shaping the creation of the Improvised Study are listed in Table 2. The aspects of compositional methods, *elements of personal interest*, are identified in the second column of Table 3 below. The third column in Table 3 provides the detail of the study developed.

In terms of *shaping factors*, particular resonances observed in the creation of the study include *theoretical, technical, practical, environmental* and *listening* elements, these listed in Table 2. Of the elements listed, the most pertinent are *technical* aspects, particularly an interest in computer interactivity, and *listening* and *environmental* aspects, including an interest in a range of musical styles and live performance. On a *micro environmental* level, the Improvised Study was realised as a live performance in a studio setting, a process common to most of the improvised works discussed herein. In terms of the practical element, *play*, my interest in creating works that involve some degree of performer autonomy reflects the concerns elaborated by Behrman and Chadabe above.

Parameter			Resonances
Musical	**Theoretical**		• University music study (all).
	Practical	**Technical**	• Computer interactivity (Behrman, Chadabe, Lewis). • Use of computer as tool which effects creation process (Chadabe). • Belief in 'amorality' of technology (Behrman).
		Play	• Long history of live performance (all). • Interest in performer autonomy, i.e., breakdown of traditional composer/performer split (all).
		Practice	
	Listening		• Acknowledged range of influential music styles (Davis, Douglas, Hancock, Oliveros, Pine). • Interest in environmental sounds (Oliveros).
Other	**Environmental**	**Macro**	• Acknowledged desire to reflect current social/culture climate, particularly popular aspects (Davis, Hancock, Pine, Oliveros). • Interest in exploring experimental means (Lewis – AACM).
		Micro	• Live performance in studio setting (all except Lewis and Interface).
	Budget/ Resources		
	Intended Audience		
	Timeframe		• Initial recording of live performance in matter of day/s (all).

Table 2. Resonances Observed Between Factors Shaping Selected Key Works and Factors Shaping the Creation of the Improvised Study

Parameter		Elements of Interest	Elements of Study
Sound	**Sources**	• Traditional instruments (all). • Combination of acoustic instruments and electric/sampled sound sources (all). • Electric pianos (Davis, Hancock, Pine). • Computer generated samples/synthesis (Behrman, Chadabe, Interface and Lewis). • Turntable/vinyl sounds/samples (Douglas, Oliveros, Pine).	• Computer generated samples/synthesis, constructed in Reason and 'remixed' in Max/MSP. • Between one and three loops of original track heard.
	Objects	• Live instrument sounds (all). • Synthesised/sampled instrument sounds (Chadabe, Douglas, Hancock, Lewis, Pine). • Range of homogenous objects (Behrman, Chadabe). • Range of heterogeneous objects (Interface, Lewis). • Sustained wide bandwidth tone throughout (Oliveros). • Noise elements, including vinyl 'hiss' (Douglas, Oliveros, Pine).	• Synthesised/sampled instrument sounds including drumkit, acoustic bass, electric piano (Rhodes), acoustic guitar played at various speeds. • Range of homogenous 'meta-objects' created by use of up to three layers of same source.
	Object processing	• Delay (Davis, Oliveros). • Filtering (Douglas, Pine).	• Variable speed playback. • Delay.

Rhythm	**Pulse**	• Constant (Davis - mostly, Chadabe, Douglas, Hancock, Pine). • No pulse (or multiple, unmetrically related) (Behrman, Interface, Lewis, Oliveros).	• Mostly constant throughout with multiple pulse often. • Main loop 173 bpm (heard at 0'1'37" and 2'43 – 3'50"). • Middle loop 133 bpm (at 1'15 – 2'08")
	Metre	• 4/4 (Davis & Hancock – mostly, Pine, Douglas). • 5/8 (Chadabe). • Variations (Davis, Hancock). • Ametric (Berhman, Interface, Lewis, Oliveros).	• Main and middle loops 2/4.
	Patterning	• Repetitive, syncopated elements (Chadabe, Davis, Douglas, Hancock, Pine). • Constant $1/8^{th}$ or $1/16^{th}$ note phrases in solos (Chadabe, Davis, Douglas, Hancock, Pine). • Free (Behrman, Interface, Lewis, Oliveros).	• Repetition and syncopation in main groove. • Repetition of rhythms throughout, with variable speed creating isometric effect.
Pitch	**Selection**	• Limited pitch set (Behrman). • Tonal with some chromatic elements (Chadabe, Douglas, Hancock, Pine). • Polytonal elements (Davis). • Microtonal elements (Lewis). • Mostly non-pitched or indeterminate pitch (Interface, Oliveros).	• Tonal, polytonal and microtonal elements.
	Vertical structures	• Chord structures and voicings based on extensions and alterations of diatonic harmony as per jazz style (Davis, Douglas, Hancock, Pine). • Somewhat random vertical alignment of pitch elements (Interface, Lewis, Oliveros). • Range of intervals from semitone to octave utilised (all).	• Chord structures and voicings based on extensions of diatonic harmony as per jazz style. However multiple loops heard simultaneously and therefore actual vertical alignment of pitched events somewhat random. • Three modes of vertical alignment heard in study and result from various playback speed formula: '(a) matching' - where computer generated playback speed (*cps*) is a factor or multiple of player playback speed (*pps*)(i.e., 0.25, 0.34, 0.5, 0.67, 0.75, 1, 1.5, 2, 3 or 4 times player speed); (b) 'opposing' – where *cps* is determined by following (if *pps* < 2, *pps* + *cps* = 2, or if *pps* >2, *pps* + *cps* = 500); or (c) 'ignoring' – where cps changes at regular intervals by semi-random amount (using Max drunk object). • Intervals vary according to playback speed. At normal speed (i.e., speed of main loop) mostly stepwise or third intervals with some larger intervals up to and including a perfect fifth.
	Vertical patterning	• Static harmony or pedal point (Behrman, Davis). • Repeating chord progression (Chadabe, Douglas, Hancock, Pine). • Random (Lewis). • Nil (Interface, Oliveros).	• One layer provides static harmony for most of work. Main riff - contrapuntal elements outline Eb Minor 7 (dorian) tonality. Middle riff (at 1'15" – 2'08") – pitches include C – Eb – G (electric piano) and Gb – Db (guitar). • Somewhat random patterning from other layers and presents polytonal/microtonal elements at times. (Three playback speed modes – see above - determine actual combinations).
	Horizontal structures and patterning	• Motive development, use of sequences, diminution, augmentation etc., (Behrman, Chadabe, Davis, Douglas, Hancock, Lewis, Pine). • Apparent random elements (Behrman, Chadabe, Interface, Lewis).	• Repetition of elements throughout with isorhythmic shifts due to changes in playback speed. • Main 'loop feature repeating Ab – Gb - Eb descending figure. • Middle riff features repeats Eb – C (descending) – G (ascending).
	Dynamics	• Shifts due to texture (all). • Fairly constant throughout (Behrman, Chadabe). • Peaks and troughs creating tension and release (Davis, Hancock, Interface, Lewis, Pine).	• Shifts due to texture and performance. • Decrescendo on main loop from 1'20" – 1'37", fade to silence. • Some fades on other layers.
	Texture	• Instrumentation includes: drumkit, bass, guitar/keyboard, sax/trumpet (Davis, Douglas, Hancock, Pine).	• Varies between 1 – 3 homogenous 'meta-layers'. Each meta-layer includes drumkit, bass, electric piano and acoustic guitar samples. (See form

		• Homogenous (Behrman, Chadabe) • Thick diverse, heterogenous (Davis, Interface, Lewis). • Major shifts marking sections or start/end of solos (Davis, Douglas, Hancock, Pine).	diagram below, Fig. 1).
	Timbre	• Mostly static timbres throughout (Behrman, Chadabe, Davis). • Foregrounding of timbral transformation (Hancock, Interface, Lewis, Oliveros). • Use of extended instrument techniques (Interface, Lewis).	• Mostly static timbres throughout with changes due to variations in playback speed.
	Spatial elements	• Placement of sounds in stereo field (all). • Blurring of source of multiple timbrally similar sounds (Chadabe). • Lack of foreground/background relationships (Behrman, Interface, Lewis, Oliveros). • Foregrounding soloists in solo/accompaniment texture (Davis, Douglas, Hancock, Pine).	• Sounds mostly centred with some spread (via reverb) across stereo field. • Delayed sound heard at far left and right. • Lack of clear foreground/background relationships with consistent elements (i.e., main and middle loops) mostly at same volume as other elements. Exception to this at beginning (0 – 40") where main loop is perceived as foreground.
	Programmatic Association	• Movement from centered to dispersed achieved by range of parameters including spatialisation, instrument roles, melodic/harmonic aspects, structure and effects (Davis). • Use of vinyl hiss adding 'warmth' to track (Douglas, Oliveros, Pine). • Random, 'non-human' feel at times (Behrman, Chadabe, Interface Lewis). • Repetitive, syncopated rhythmic aspects providing somatic/dance aspect (Davis, Douglas, Hancock, Pine).	• Juxtaposition of constant and varying elements (echoing centred/dispersed movement of Davis) created by maintaining one layer (main loop or middle loop) for most of work. • Random nature of changes in playback speed combined with 'skipping' feel of loops create non-human feel. • Repetitive and syncopated aspects provide somatic/dance aspect albeit at very fast tempo and with somewhat 'unhuman' feel.
	Structure	• Sectional with solo/accompaniment sections (Chadabe, Davis, Douglas, Hancock, Pine). • Free (Behrman, Interface, Lewis, Oliveros).	• Sectional: ABA Coda, with sections determined by presence of main (A) or middle (B) loops. Coda without main/middle loop. See form diagram below. • Length of sections freely determined during performance. • Main and middle loops not precomposed or predetermined, resulting instead from chance placement of start and end points on sample.
	Interaction	• Solo/accompaniment sections with some accompaniment parts 'locked' (Davis, Hancock, Pine, Douglas). • Collective improvisation sections between two or more sources (all). • Blurring of role between soloist/accompanist (Behrman, Chadabe, Davis, Lewis, Oliveros). • Non-hierarchical environment (Behrman, Lewis). • On timbral and textural levels (Interface, Lewis). • Computer interaction (Behrman, Chadabe, Lewis). • Computer interaction based on rules – match, oppose or ignore (Lewis). • Tension and release principle achieved via variety of parameters, e.g. pitch, dynamics, rhythm, texture, and/or structure (all).	• Computer interaction system developed in Max/MSP. Multiple sample (one to four) playback with computer controlling playback speed of two samples. Rules based on 'match', 'oppose' or ''ignore' principles (Lewis). However player controls other parameters of computer playback (e.g., interaction mode, sample selection, volume and loop start and end point selection). • Level and nature of interaction varies between sections. See form diagram (Fig. 1) for graphic representation. • 'Matching' and 'ignoring' modes of interactive system utilised. • 'Locking' of some parts (e.g., main and middle loops). • Tension and release achieved by: removal and return of main loop; alternating between 'matching' and 'ignoring' modes of computer (e.g., 2'58" – 4'22").

Table 3. Elements of Personal Interest and Elements of Study

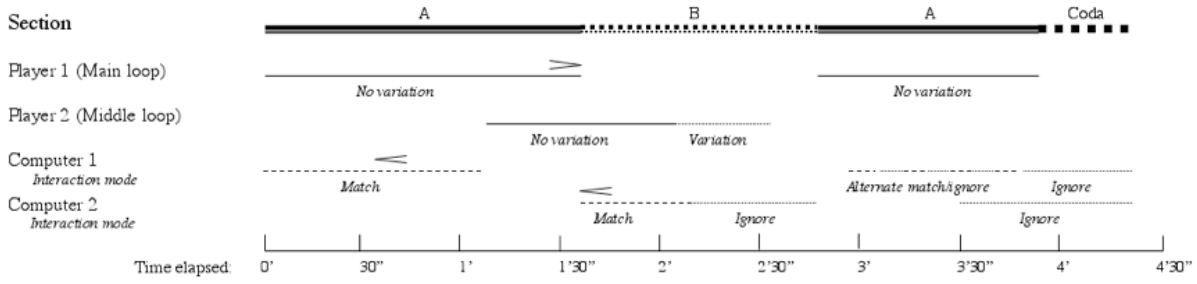


Figure 1. Improvised Study: Form diagram

In terms of *inputs* for the study, many of the *elements of interest* (listed in the second column of Table 3) were incorporated into the study (listed in the third column of Table 3). This extends to the incorporation of somewhat disparate elements of interest within individual parameters. For example, the *sound source* utilised for the study is a draft of an instrumental work for drumkit, electric piano, double bass and guitar that was realised using samples within the Reason software application. For the study, multiple short sections of this work are sampled and played at various speeds. Thus, in some way, all of the sound sources listed as elements of interest in Table 3 are incorporated into the study. Samples of traditional instruments, including electric piano, are replayed in the study in a manner echoing the use of a turntable as source in both Oliveros' "Bye Bye Butterfly" and Pine's "Oneness of Mind". Furthermore, as in the Pine example, samples of my own performance are used as a sound source.

Other examples of the incorporation of disparate elements of interest into the study include elements of *rhythm* and *pitch*. In terms of rhythm, for the works analysed, two categories can be established; (a) works with a constant pulse and (b) works without or with multiple or unmetrically related pulse. For the study both categories are explored by foregrounding a constant pulse in some sections and overlaying irregular/unmetrically related fragments at other times. In terms of *interaction*, I was particularly interested in the interactive computer systems developed by Behrman, Chadabe and Lewis. An integral component of all of these systems is analysis and computer performance on a note-to-note level, and the incorporation of some traditional pitch elements. However, given my ongoing interest in utilising field recordings and sound fragments above and/or below note level in terms of timescale the study incorporates computer interactivity at a conceptual level, not a note-to-note level. Instead, the study is an attempt to incorporate computer interaction within a system of multiple sample playback modules in an improvised performance context.

REALISATION OF IMPROVISED STUDY

The performance system developed in Max/MSP for the Improvised Study extends from my previous work with field recordings where various parameters such as start and end points of loop, playback speed and spatialisation of various sound fragments, between approximately three and thirty seconds in length, were manipulated in real time. The Max/MSP abstractions utilised were further developed for performances throughout 2004 and 2005 with the interactive aspects constituting the new component developed for the Improvised Study. For the study I sought to explore the computer automation of playback speed in terms of the rules of interaction utilised by Lewis in "Voyager", i.e., the three modes, matching, opposing or ignoring. However, whilst the system developed for the Improvised Study could be utilised in a 'non-hierarchical' (Lewis, 2000a) manner, my own exploration of the system involved some degree of control of the computer's output. This included at a basic level, sound source selection and volume control. Figures 2 and 3 present the basic audio and interactive control flow charts for the abstractions developed in Max/MSP, with key objects indicated where appropriate.

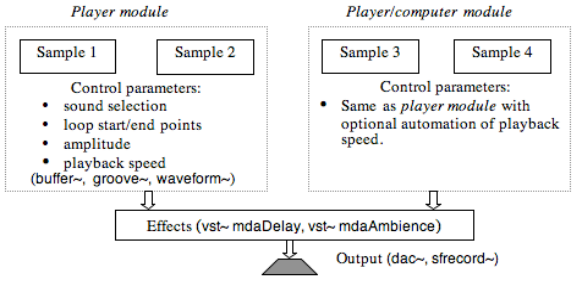


Figure 2. Improvised Study: Basic audio and control flow chart

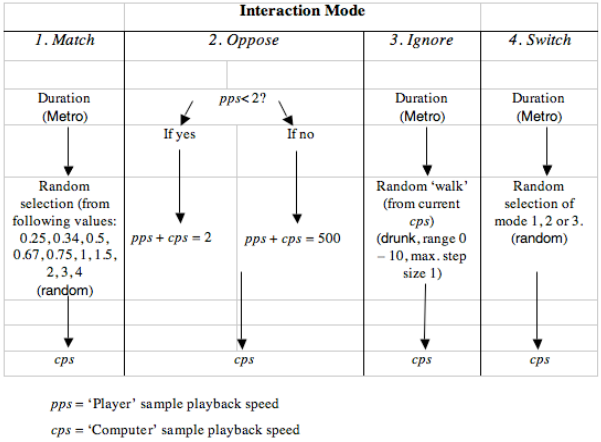


Figure 3. Improvised Study: Interaction mode flow chart

Given that the system developed was an extension of my earlier work, minimal system testing was required. However, once operational, a lengthy play process was undertaken where a range of sound sources, interaction modes and performance techniques were explored. Numerous performances were recorded and reviewed, most featuring the utilisation of only one interaction mode. Through this initial process of minimisation where, for example, only the matching or opposing modes of interaction were utilised, a more thorough exploration of the various modes and potential sonic outcomes was possible. Having established some fruitful performance avenues, further reflection on the various compositional methods identified through the analysis of the selected key works was conducted before completing the study. The Improvised Study was selected from numerous performances conducted in a home studio environment, monitoring with headphones and utilising the computer mouse, trackpad and computer keyboard for control. The study was recorded directly within Max/MSP and transferred to Pro Tools for normalisation with no further editing or effects added.

CONCLUSION

Having completed the analyses discussed herein and another twenty-seven in other genres, refinement of the analytical template is recommended by subsuming *timbre* within the description of individual *sound objects* and an expansion of *programmatic association* to include sub-parameters such as *gestural reference* (following Middleton, 2000), *signification/connotation* and *intrinsic/ contextual recognition* (following Wishart, 1996). In addition to the desk-bound method presented here, an ethnographic approach could also be used to probe collaborative processes. The relationships that develop within a studio environment between musicians, engineers and producers warrant detailed study in order to establish a thorough picture of the nature of music creation.

The analysis/creation model developed for this research makes explicit the connection between musical analysis and music creation; a connection not made in the majority of existing analytical methods. The compartmentalisation of musical inputs as discrete elements enabled the creation of a storehouse; an accumulation of discrete elements (i.e., compositional methods) at/from which I could ponder, select, combine or appropriate in the creation of the new practical works. The relative fixity of the shaping factors elements highlight individual agency and offer a starting point for consideration of the impact of such factors as gender, class and race on music creation.

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